IOT SDN NFV

Computer Vision

6_9_2_PROTOCOL LAYERING

https://github.com/siagianp
https://www.youtube.com/@AmelOline/videos
https://github.com/amelcharolinesgn2/IoT_simulator-mqtt-NodeRed
https://github.com/amelcharolinesgn2/Cloud-Incomastructures
https://github.com/amelcharolinesgn2/ClouD-Incomastructure-SISKA-2025

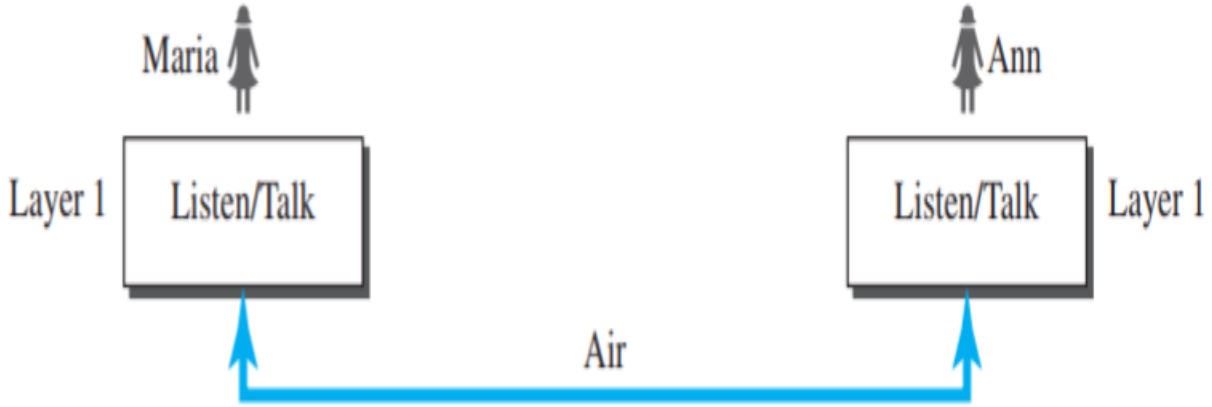
Introduction

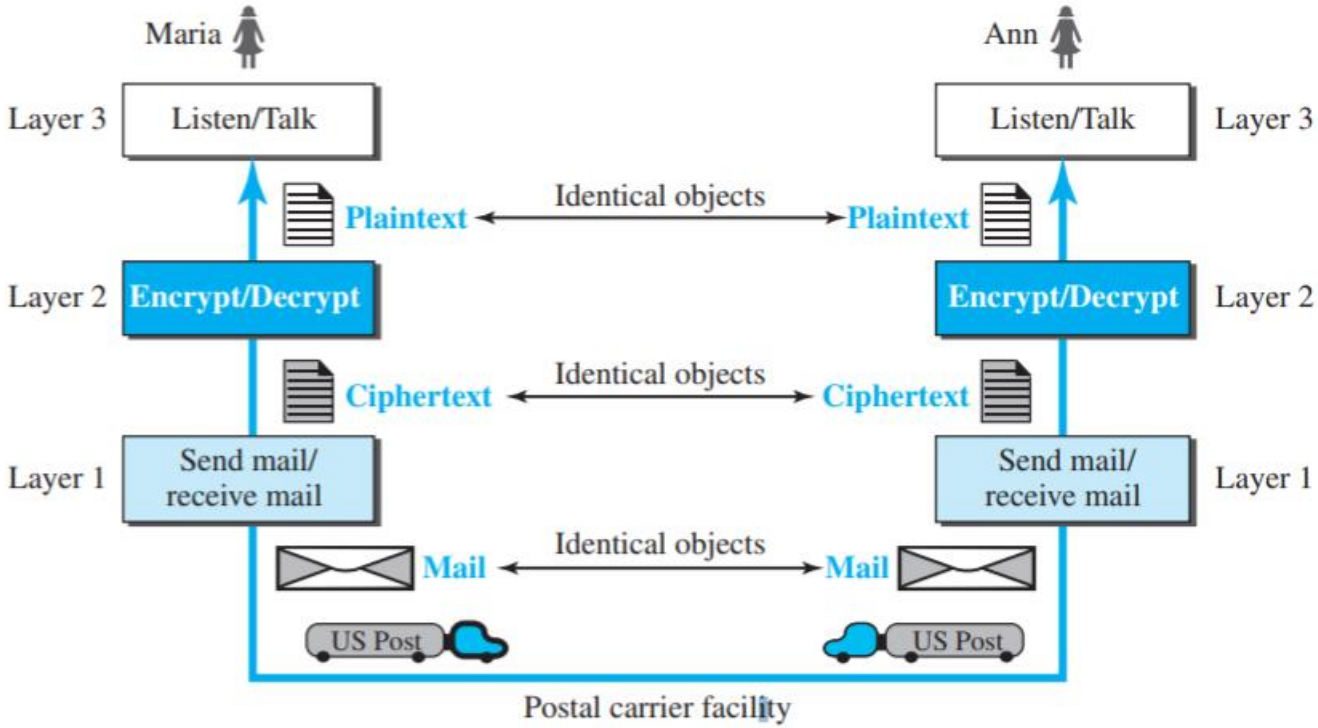
- In data communication and networking, a protocol defines the rules that both the sender and receiver and all intermediate devices need to follow to be able to communicate effectively.
- When communication is simple, we may need only one simple protocol;
- when the communication is complex, we may need to divide the task between different layers, in which case we need a protocol at each layer, or protocol layering.

To understand the need for protocol layering

Scenarios

Figure 2.1 *A single-layer protocol*




Maria
Ann
Layer 3
Listen/Talk
Listen/Talk
Layer 3
Plaintext
Identical objects
Plaintext
Layer 2
Encrypt/Decrypt
Encrypt/Decrypt
Layer 2
Ciphertext
Identical objects
Ciphertext
Layer 1
Send mail/
receive mail
Send mail/
receive mail
Layer 1
Mail
Identical objects
Mail
US Post
US Post
Postal carrier facility

What are layered protocols?

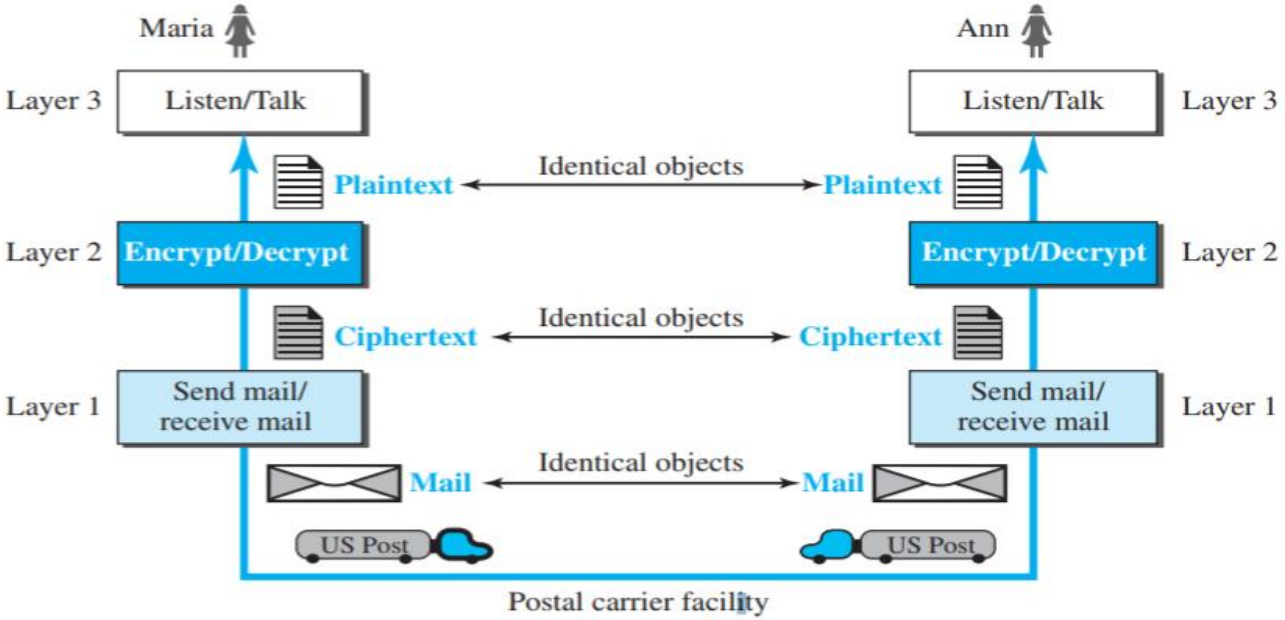
- A **layered protocol** architecture provides a conceptual framework for dividing the complex task of exchanging information between remote hosts into simpler tasks.
- Each **protocol layer** has a narrowly defined responsibility.
- P**rotocol layer** provides a standard interface to the next higher **protocol layer**.

Why are network protocol layers important?

- In order for computers with different hardware and operating systems to be able to communicate effectively over a **network** or an **internet**,
- it is clearly **important** for there to be a uniform set of **protocols** and standards which the communicating systems and applications will conform to.

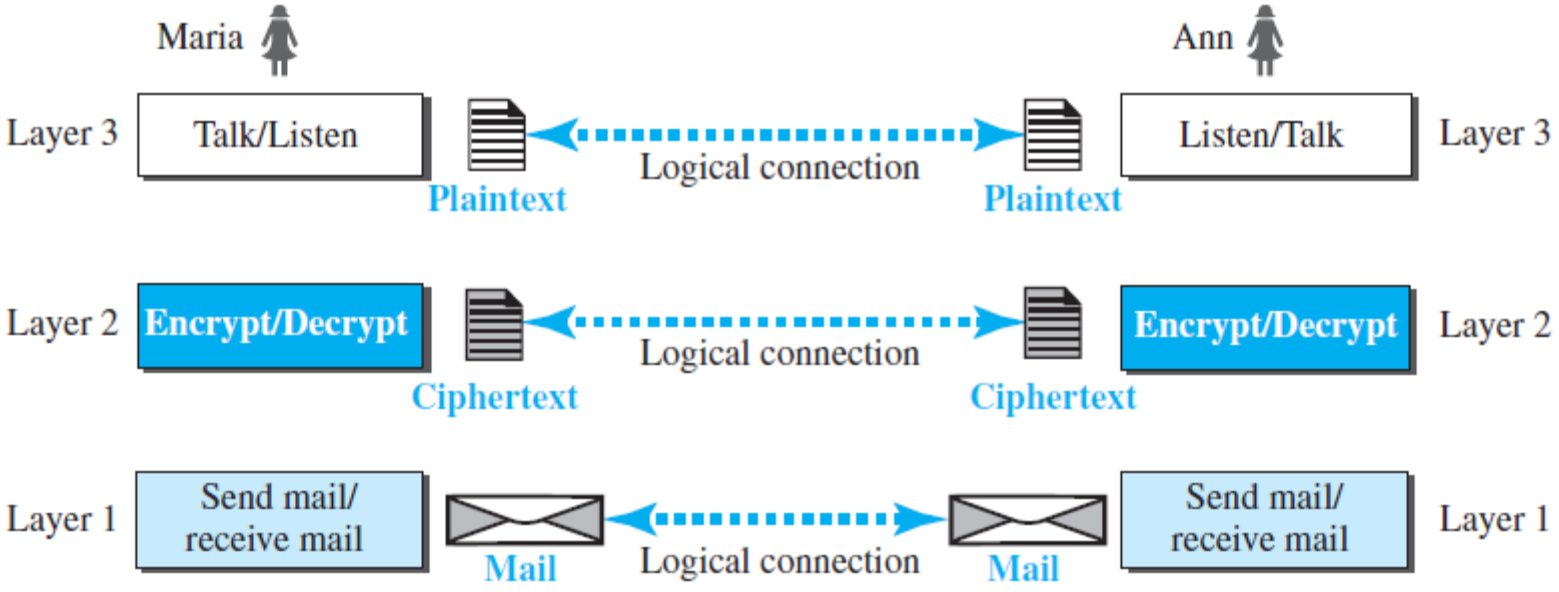
What are the two principles of protocol layering?

- The first principle dictates that if we want bidirectional **communication**, we need to make each layer so that it is able to perform two opposite tasks, one in each **direction**.
- The second principle that we need to follow in protocol layering is that the two objects under each layer at both sites should be identical.



Logical Connections

- Maria and Ann can think that there is a logical (imaginary) connection at each layer through which they can send the object created from that layer.



Protocol "layers" and reference models

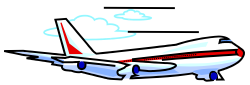
Networks are complex, with many "pieces":

- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Question: is there any hope of *organizing* structure of network?

- and/or our *discussion* of networks?

Example: organization of air travel



end-to-end transfer of person plus baggage

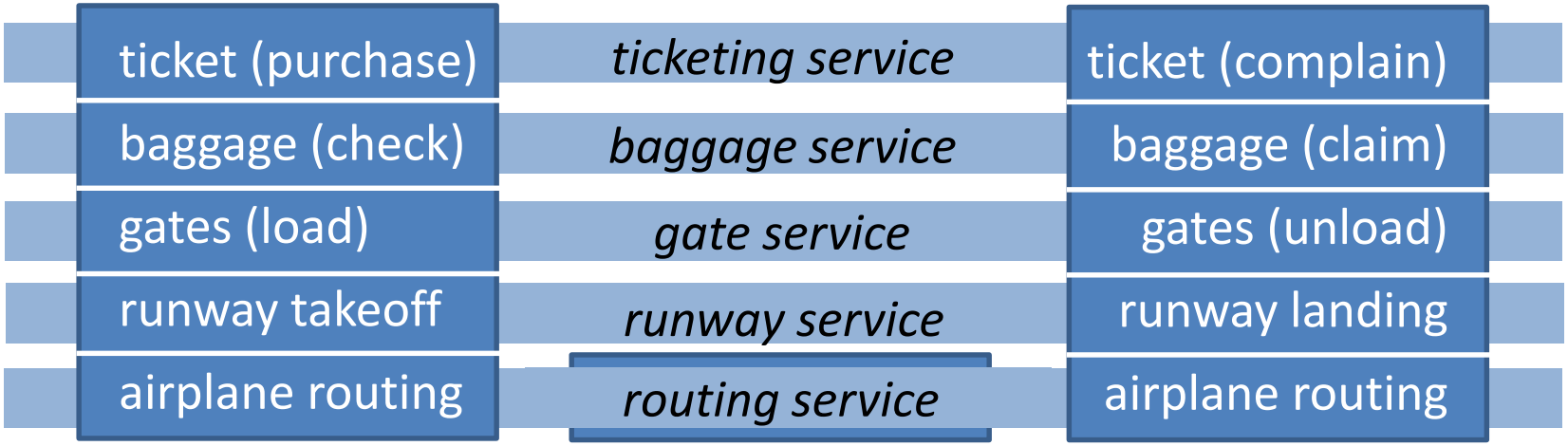
ticket (purchase)	ticket (complain)
baggage (check)	baggage (claim)
gates (load)	gates (unload)
runway takeoff	runway landing
airplane routing	airplane routing

airplane routing

How would you *define/discuss* the *system* of airline travel?

- a series of steps, involving many services

Example: organization of air travel



layers: each layer implements a service

- via its own internal-layer actions
- relying on services provided by layer below

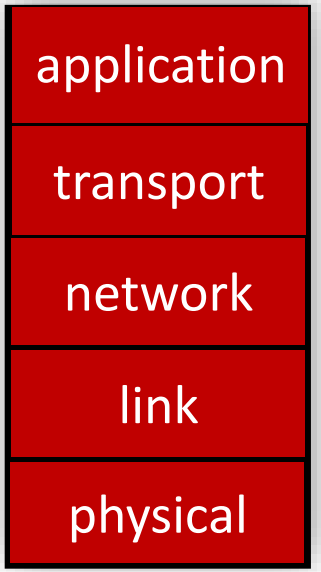
Why layering?

Approach to designing/discussing complex systems:

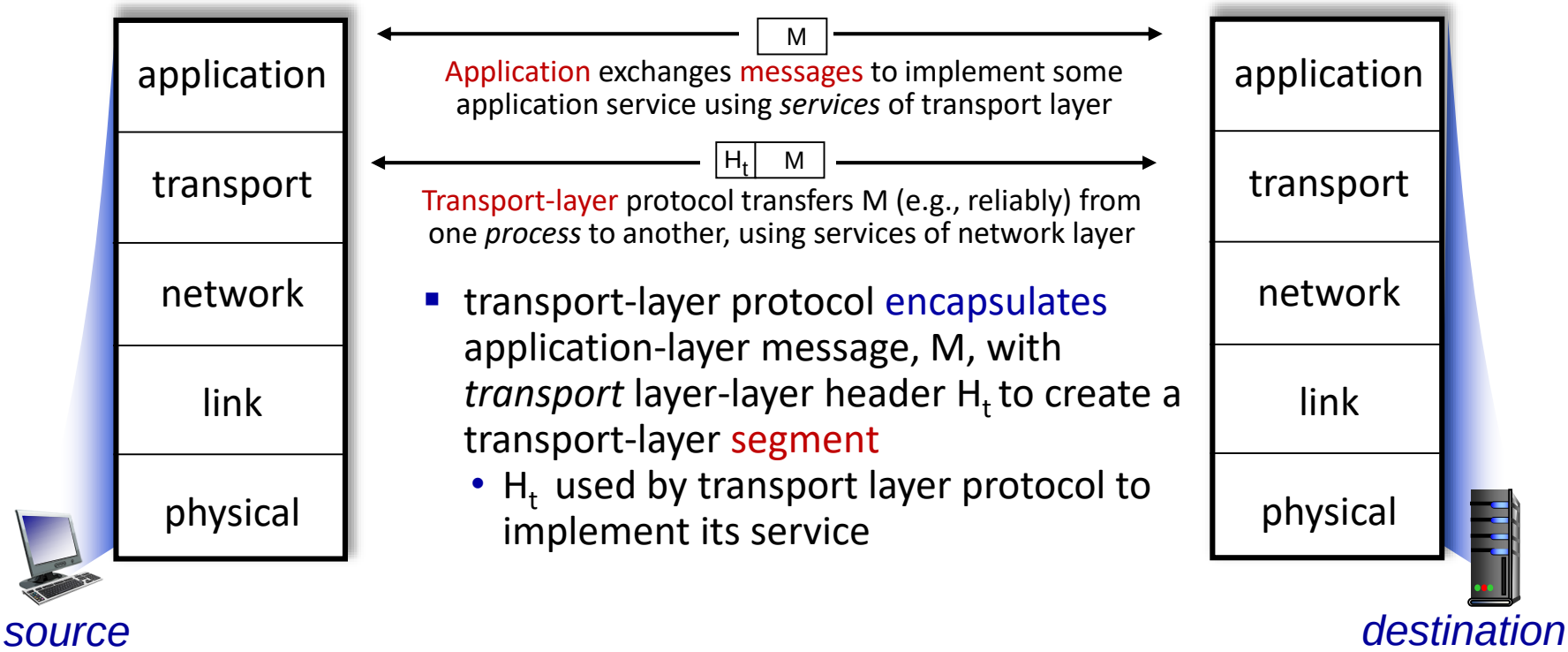
- explicit structure allows identification, relationship of system's pieces
  - layered *reference model* for discussion
- modularization eases maintenance, updating of system
  - change in layer's service *implementation*: transparent to rest of system
  - e.g., change in gate procedure doesn't affect rest of system

Layered Internet protocol Stack

- *application:* supporting network applications
  - HTTP, IMAP, SMTP, DNS
- *transport:* process-process data transfer
  - TCP, UDP
- *network:* routing of datagrams from source to destination
  - IP, routing protocols
- *link:* data transfer between neighboring network elements
  - Ethernet, 802.11 (WiFi), PPP
- *physical:* bits “on the wire”

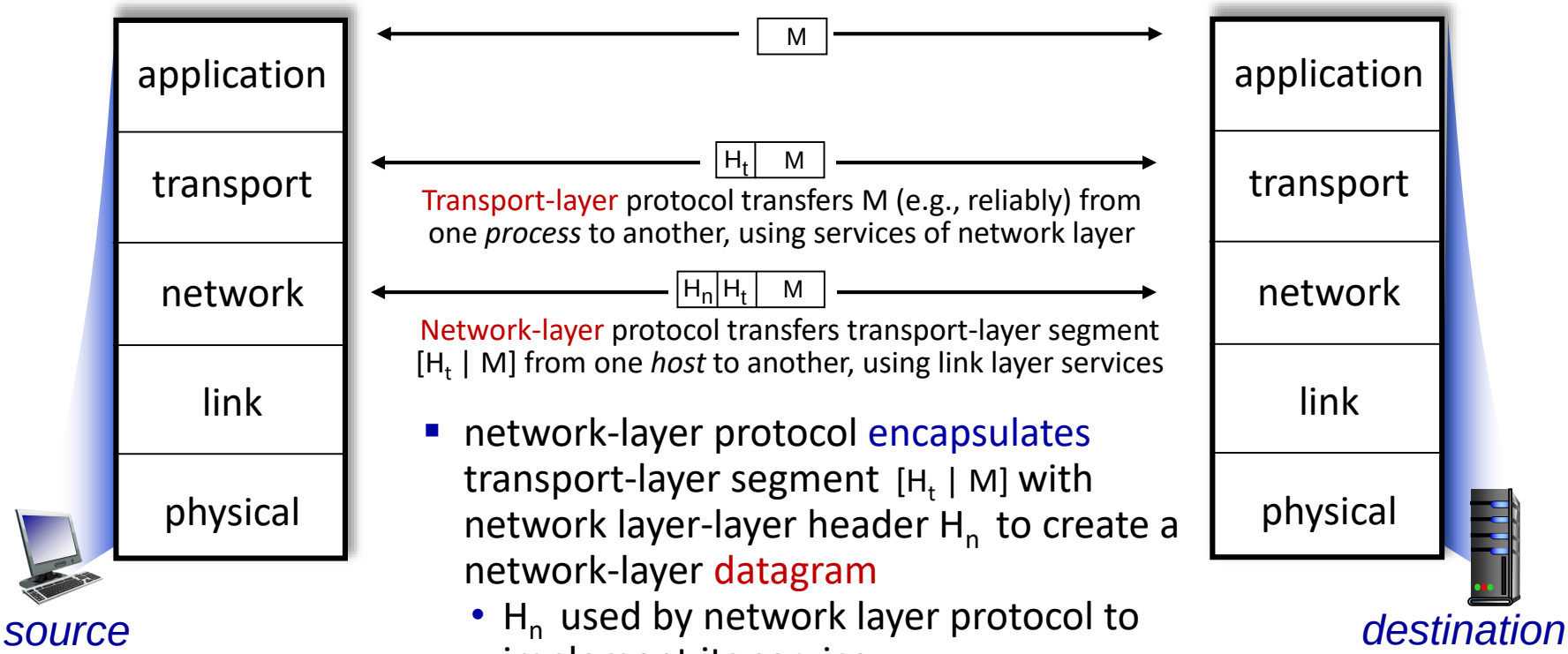


Services, Layering and Encapsulation



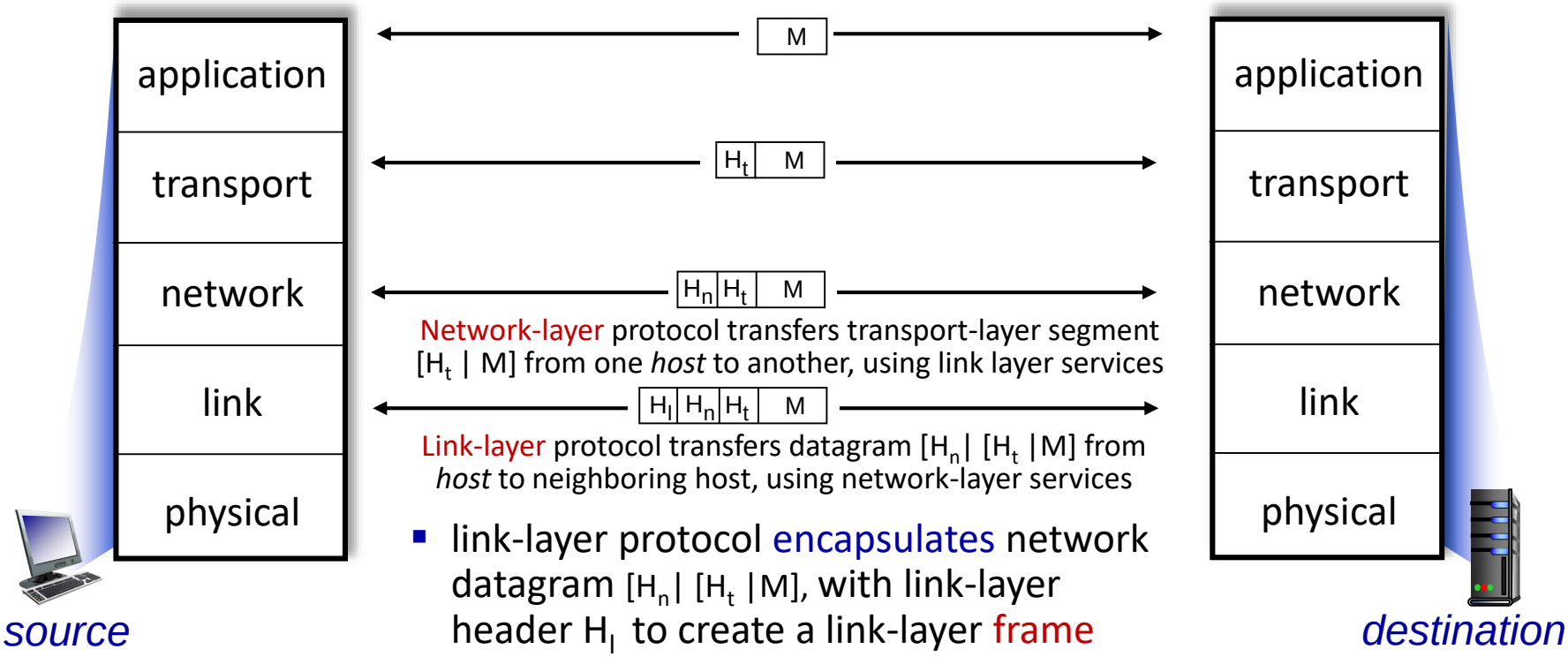
- transport-layer protocol encapsulates application-layer message, M, with *transport* layer-layer header H_t to create a transport-layer segment
  - H_t used by transport layer protocol to implement its service

Services, Layering and Encapsulation



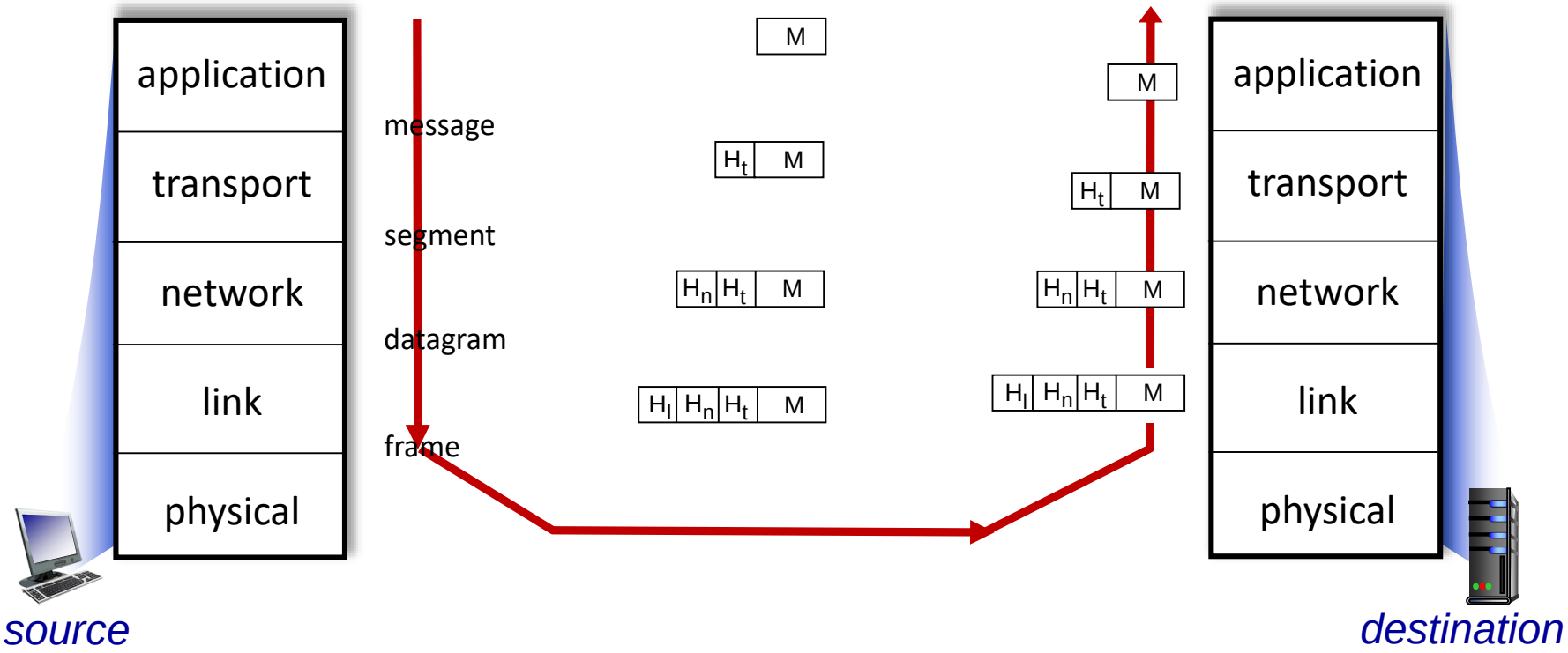
- network-layer protocol encapsulates transport-layer segment [H_t | M] with network layer-layer header H_n to create a network-layer datagram
  - H_n used by network layer protocol to implement its service

Services, Layering and Encapsulation

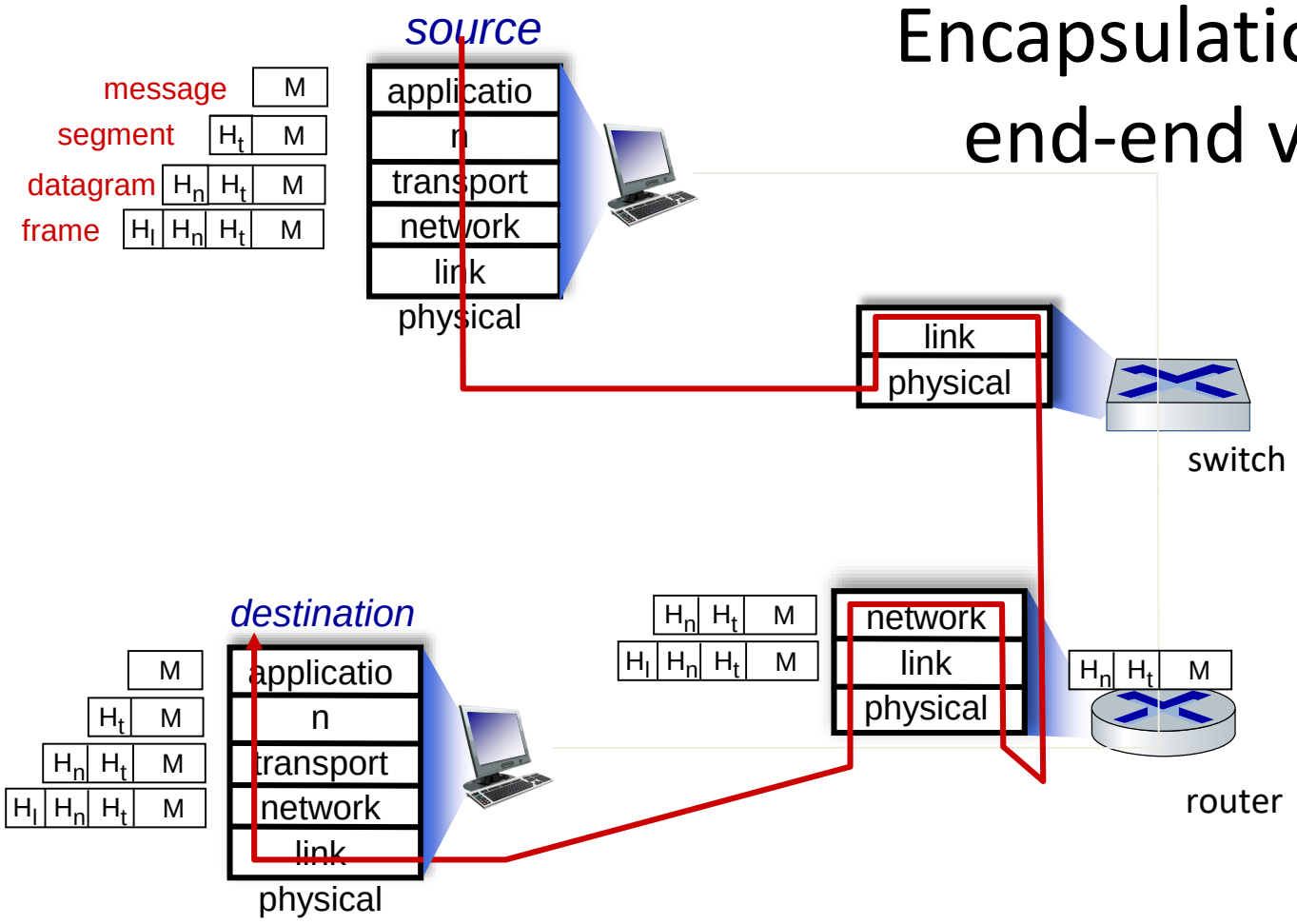


- link-layer protocol encapsulates network datagram [H_n| [H_t |M], with link-layer header H_l to create a link-layer frame

Services, Layering and Encapsulation



Encapsulation: an end-end view



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Thank You !!

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